

GPS Module Features

Feature	Benefit	Effect of increase	Effect of decrease
Size	Affects antenna choice and features, and is based on application size	Bigger modules allow for a bigger antenna, which gives better lock time and accuracy	Smaller modules are used more in wearable electronics and mobile computing where size matters
Update rate	Speed with which the module recalculates and reports NMEA data	High frequencies allow the module to keep up with higher speeds, like when flying	Lower frequencies allow the module to work with low power and reduced computing strain
Power	Depends on the performance expectation that the device is tuned for	Higher update rates and bigger antennae can be used if lots of power is available	Low update rates. If the power figure is really low, you probably forgot to connect the antenna
Accuracy	Depends on the accuracy demanded by the application	High-accuracy modules are required for highly critical applications such as military and aerospace	Low accuracy is perfect for home use as it is inexpensive and decreases BoM
Antennae	Fundamental right of a GPS module	Bigger antennae need more power to be driven to them by the LNAs	Smaller antennae are usually inexpensive and also fit for devices bound by size limitation
Channels	Channels are how the module would receive ephemeris data from the satellite	More channels mean more power consumption but a faster TTF	Fewer channels increase TTF a lot, one solution is to cut down on channels after first fix
Extras	New modules come with jamming detection and mitigation	High sensitivity of even -156 dB, and availability of 60+ channels	Magnetic sensors and accelerometers might be available to aid GPS lock

the other hand, passive antennae are directly mounted on the PCB and are more susceptible to noise coupling if not correctly isolated. There are also more expensive high-performance antennae that are bigger and more power consuming but deliver exceptional accuracy.

For precision applications requiring resolution in the millimetre range, it is paramount that signals from satellites at all heights virtually meet at exactly the same point inside the antenna. For this type of application, receivers with multiple antenna inputs would be required.

For most applications, it is easier to buy a GPS antenna module from vendors like SANAV, Johanson Technology, Fastrax or GlobalTop. The choice of component varies depending on a lot of parameters like required attenuation, power consumption, size and connectors.

Application focused. One development related to the *Electronics of Things* concept is the fusion of sensor data by combining together data from multiple

sensors in the device, like accelerometers, gyros and magnetic sensors.

“Dynamic contextual awareness, temperature monitoring and MEMS sensors work in concert to conserve power and boost performance. This opens up a wide variety of use cases for sports camera, asset tracking and fitness devices,” explains Reuven Samuel, senior product manager, location marketing, CSR Sunnyvale.

Technology like dead reckoning aids GPS navigation based on algorithms that consider distance and directional changes to boost GPS accuracy even when a GPS signal is not available. This is currently seen in some of the latest automotive navigation systems.

When consolidating multiple functionalities into a single system in a typical IoT concept, the biggest challenge from an operating system perspective is to be able to meet the resource demands of the various applications running on top of it.

“Multiple applications that at one point ran on their own set of hardware

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